

Choosing the Right Machine Learning Model When Your Data is Imbalanced

5 min read · Apr 3, 2026



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So let's say we want to build an AI that can understand human gestures for, say, sign language translation or human-computer interaction. To do this, we would need to acquire the correct data to run it. The data would unfortunately have some movements occurring far more than others, This reflects in a real-world imbalanced gesture data. If the method/algorithm which runs your system isn't careful, it will ignore the rare but important ones, making your AI redundant. In this article, I am trying to investigate how to thoughtfully select a machine learning classifier for exactly this kind of problem, using the Gesture Phase Segmentation dataset.

Understanding the Problem and the Data

We want to automatically detect and segment different phases of human gestures from motion sensor data like Kinect. The dataset in use, called the Gesture Phase Segmentation dataset contains about 9900 instances and 32 features. In essence, it contains things like hand position, wrist velocity, head acceleration, etc. The goal is to classify each moment into one of five natural gesture phases:

1. Preparation
2. Stroke (some meaningful action/movement)
3. Hold
4. Retraction

5. Rest



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(a) Preparation (b) Stroke (c) Hold (d) Retraction (e) Rest

A subset of gestures

Like in the real world, 'Rest' and 'Stroke' obviously appear very often than 'Hold' and 'Retraction'. If your model keeps missing the latter two underrepresented phases, the entire meaning of the gesture sequence can break down.

Choosing the Right Success Evaluation Metrics

To remedy this imbalance, you need to learn about some metrics which when focused on could indicate the correct handling of these underrepresented phases. That should help decide how you'll measure success of your model selection.

Accuracy alone is misleading here because of the imbalance. Instead better choices include:

1. Balanced Accuracy, which gives equal importance to every phase
2. Macro F1 Score, which treat all five classes fairly
3. Macro Recall, which is especially useful when missing rare classes are impactful

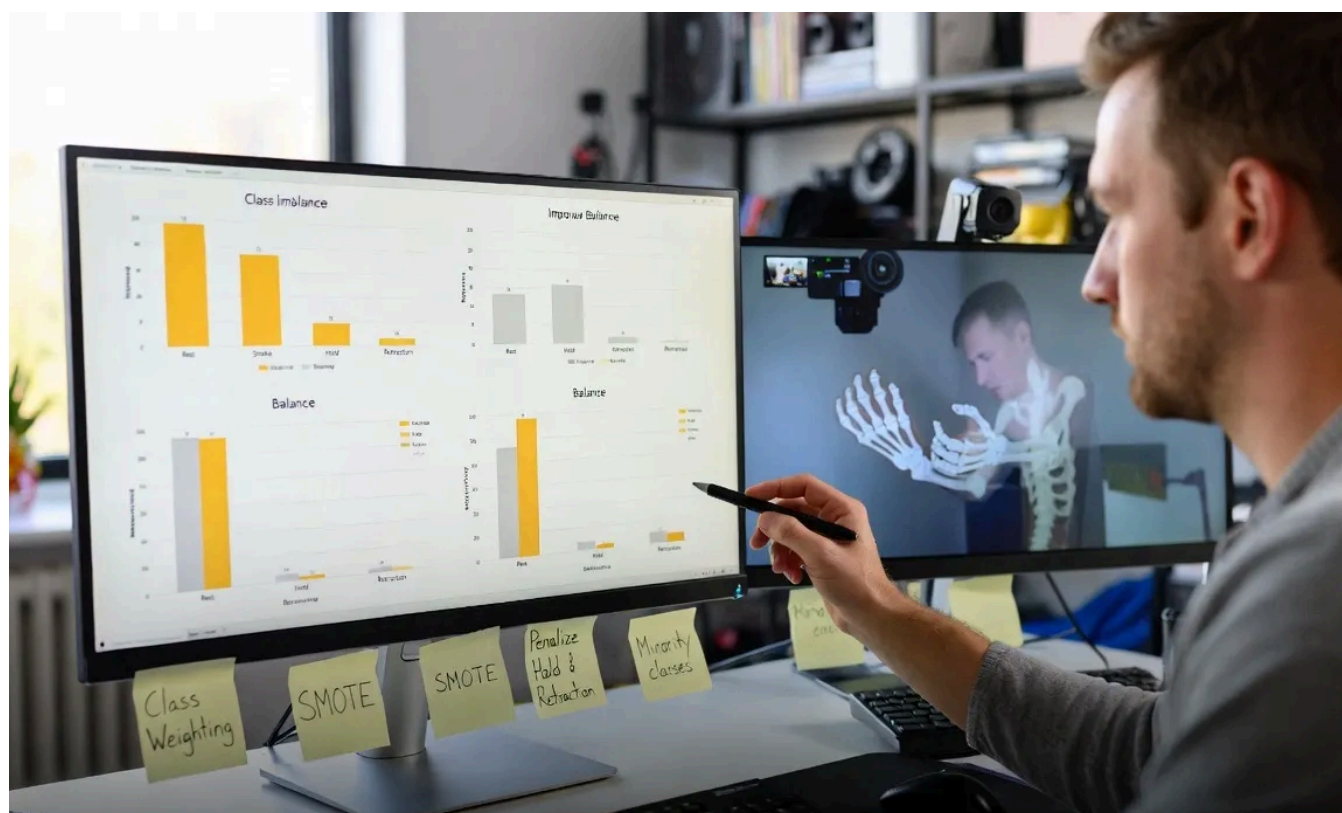
4. AUC (Area Under the Curve), which helps evaluate how well your model ranks predictions/discriminates among the phase classes

Preparing the Data

Two important steps make a big difference given the nature of our data. They are:



Applying Class Weighting: Instead of letting the model lazily predict the common classes ('Rest' and 'Stroke'), we can penalise it more heavily when it misclassifies the rare ones ('Hold' and 'Retraction'). This I believe, is one of the most effective ways to address class imbalance.



Handling class imbalance

Feature Scaling: If distance-based methods like the K-Nearest Neighbours (KNN) are considered, it becomes important to scale your features. This prevents features with larger numbers from unfairly dominating the model's decisions.

Exploring Different Models

Let the modelling begin! It's always a good idea to establish a strong baseline to compare other models with.

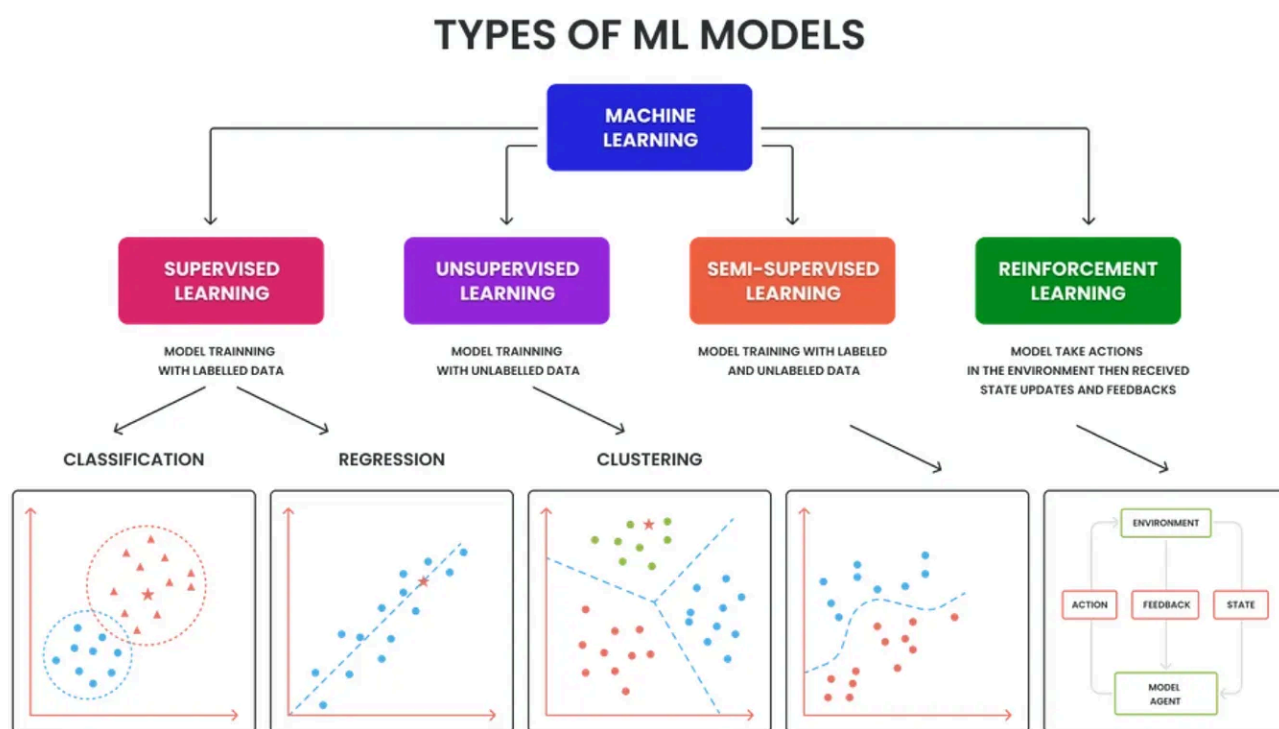


Image from [Aristek Systems](#)

Random Forest

The Random Forest to me could be that model, because its characteristics suits the nature of the dataset. It is an *ensemble* model, which means it builds many decision trees and combines their opinions to reach a final answer. You can think of it as consulting a panel of advisors rather than relying on ones. It handles non-linear relationships within some data well, works with imbalanced data. It provides probability scores, which tell you which features are the most important.

The K-Nearest Neighbours (KNN)

I can imagine the algorithm as asking your nearest neighbours for advice. Its simply looking at the most similar past examples to go ahead with the majority vote. I think it is effective when local patterns matter and it handles multi-class problems naturally.

Support Vector Machine (SVM)

The SVM attempts to draw the widest possible boundary between classes. I think it is a remarkably precise boundary-finder. Its setup is designed for 2-class datasets, since its a binary classifier at heart, but you can interestingly adopt them to handle our five-class problem.

Gradient Boosting

One of the strongest modern classifiers. Random Forest builds many trees independently and averages them, but Gradient Boosting builds them in sequence, with each new tree learning from the mistakes of the one before it. The result is a model that gets progressively better at the hard cases, including the minority classes.

So which Model Should You Actually Choose?

After training and evaluating these models on the Gesture Phase Segmentation dataset, two models stood out:

1. **K-Nearest Neighbors (KNN)** emerged as the most balanced classifier across all five gesture phases. It demonstrated the greatest fairness across all five gesture phases, making it particularly effective at avoiding bias toward the more frequent phases.
2. The **Random Forest**, in contrast, proved to be the strongest overall performer, particularly in its discriminative power, AUC scores, and probability calibration. It excelled at ranking predictions with high confidence.

Ultimately, the best model would depend on your specific use case of your AI system rather than a single “best” algorithm. The key is aligning the model choice with the real-world requirements of your application:



Aligning your model choice to your pressing business need. Image by the [Strategy Institute](#)

- If your AI requires **fair and equitable selection/detection** across all gesture phases, especially ensuring the minority classes (Hold and Retraction) are reliably detected, the **KNN** would be the recommended choice.
- If your application depends on **strong probability rankings or confidence scores** (for example, in decision-making systems that threshold predictions), **Random Forest** is the superior option.
- Otherwise to **aggressively address class imbalance** and minimise costly errors on the rare phases (Hold and Retraction), **Gradient Boosting** (such as XGBoost or LightGBM) should be given strong consideration.

Final Thoughts

In the end, no model is universally best. The right question is not “Which model performs best?” but rather: “**Which model best serves the specific needs and risks of this application?**” Success of your gesture recognition system will ultimately come down to how well the chosen model aligns with your true objectives, the cost of different types of errors, and a deep deep understanding of the data.



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